GRAMVERSE AI

A Gamified 3D Digital Twin for Participatory and Data-Driven Rural Development Planning

Smart India Hackathon 2024 — Problem Statement SIH1704
Gamification for Rural Planning using Drone Land Survey Maps and GIS Data

1. Executive Brief

GramVerse AI is an interactive rural planning platform that transforms village-level GIS and drone-survey data into a navigable 3D digital twin. Instead of examining static maps and disconnected datasets, users can explore a virtual representation of a village, identify infrastructure gaps, propose development projects, allocate a limited budget, simulate outcomes, and compare multiple planning scenarios.

The central idea is simple: **before spending money in the real village, stakeholders should be able to test development decisions digitally.**

Gamification is used seriously—not through cartoon points or meaningless rewards, but through scenario-based missions, budget constraints, measurable outcomes, trade-offs, and planning scores.

2. Problem Statement Interpretation

SIH1704 focuses on using drone land-survey maps and GIS data for rural planning, with elements such as village selection, building footprints, road networks, 3D visualization, planning of development activities including roads and drainage, budget considerations, and improved realism.

A weak interpretation would be to build only a GIS dashboard or a village-building game. GramVerse combines the required elements into a planning workflow:

> **Understand the village → identify a gap → create a development plan → simulate the impact → compare alternatives → recommend a better allocation of resources.**

3. Core Problem Being Solved

Rural development decisions involve competing priorities. A limited budget may need to cover roads, drainage, water access, public facilities, or connectivity improvements. The difficult question is not merely what can be built, but which combination of projects provides the greatest benefit.

• Planning information can be difficult for non-technical stakeholders to interpret.

• Static maps do not easily communicate the consequences of alternative decisions.

• Limited budgets force trade-offs between multiple development needs.

• Different planning scenarios may benefit different populations or areas.

• Stakeholders need a way to compare options before implementation.

4. Proposed Solution

GramVerse creates a digital twin of a village from available geospatial layers. Users interact with the model and create planning scenarios. Each proposed intervention changes the analytical model and generates measurable outcomes.

Example interventions:

• Construct or upgrade a road

• Improve drainage

• Add or improve access to a public facility

• Improve water-related infrastructure

• Prioritize an underserved zone

• Allocate a fixed development budget across multiple interventions

5. User Journey

Step 1 — Load Village Data
The system imports GIS layers such as building footprints, roads, public assets, water bodies, and other available spatial information.

Step 2 — Generate Digital Twin
The 2D spatial information is converted into an interactive 3D representation of the village.

Step 3 — Analyze Existing Conditions
The system calculates accessibility, infrastructure coverage, spatial gaps, and other selected indicators.

Step 4 — Select a Planning Challenge
The user receives a realistic scenario with a budget, objectives, and constraints.

Step 5 — Create a Development Plan
The user places proposed interventions on the map and allocates resources.

Step 6 — Run Simulation
The platform recalculates relevant metrics after the proposed changes.

Step 7 — Compare Scenarios
Multiple plans can be compared side-by-side using cost, beneficiaries, accessibility, infrastructure improvement, and sustainability indicators.

Step 8 — Generate Recommendation
An optimization/recommendation engine identifies high-impact alternatives and explains the trade-offs.

6. System Workflow

DATA INPUT
GIS layers + road networks + building footprints + optional drone imagery
↓
DATA PROCESSING AND STORAGE
Spatial database and preprocessing
↓
3D DIGITAL TWIN
Interactive village visualization

↓
GAP AND ACCESSIBILITY ANALYSIS
Identify underserved or low-connectivity zones
↓
USER / AI DEVELOPMENT PLAN
Roads, drainage, facilities, budget allocation
↓
SIMULATION AND OPTIMIZATION
Recalculate network and impact metrics
↓
SCENARIO COMPARISON
Compare multiple plans
↓
RECOMMENDED DEVELOPMENT STRATEGY

7. Core Modules

A. GIS Data Engine
Imports and manages spatial layers using formats such as GeoJSON. The system should remain data-source agnostic so that the prototype can work with sample data while supporting future official datasets.

B. 3D Village Digital Twin
Transforms geographic features into an interactive 3D environment. Buildings, roads, water bodies and proposed interventions can be visualized spatially.

C. Infrastructure Gap Analysis
Calculates selected indicators such as distance to roads or facilities, connectivity, and coverage to identify areas requiring attention.

D. Planning Sandbox
Allows users to place interventions and experiment with different development strategies.

E. Budget and Constraint Engine
Tracks available resources and prevents invalid plans that exceed constraints.

F. Simulation Engine
Updates the model when infrastructure changes and recalculates relevant metrics.

G. Optimization Engine
Searches for better allocations of limited resources using graph algorithms and optimization logic.

H. Scenario Comparison Dashboard
Compares user-created and AI-suggested plans using transparent metrics.

I. Gamified Challenge Layer
Provides missions, constraints, objectives and scores to make planning participatory and easier to understand.

8. The Algorithmic Core

One of the strongest technical components can be accessibility analysis. The village road network can be represented as a graph:

Nodes: intersections, households or important locations
Edges: roads or paths with distance, travel time, condition and potential upgrade/construction cost.

When a user proposes a new road or upgrade, the graph changes. Shortest paths and accessibility metrics can then be recalculated. This creates a direct connection between a visible 3D change and a measurable planning outcome.

Potential techniques include shortest-path algorithms, network centrality, coverage analysis, multi-objective optimization and priority scoring.

9. AI: Where It Should and Should Not Be Used

Good AI/ML applications:

• Computer-vision-assisted extraction of roads, buildings or water bodies from drone/orthophoto imagery

• Prediction or estimation of infrastructure demand when appropriate data exists

• Spatial clustering to identify underserved regions

• Recommendation models based on historical or simulated planning scenarios

Important: The core MVP must not depend entirely on a complex ML model. A strong hybrid system using GIS analysis, rule-based scoring and optimization is more feasible and easier to defend in a hackathon.

10. Gamification Design

Gamification should reinforce planning decisions. Example missions:

• **Flood Resilience Challenge:** Improve drainage and connectivity within a fixed budget.

• **School Access Challenge:** Maximize the number of households with efficient access to a school.

• **Sustainable Village Challenge:** Improve infrastructure while minimizing environmental impact.

• **Budget Crisis Challenge:** Achieve the highest development score after a sudden budget reduction.

The score can combine weighted indicators such as population benefit, accessibility improvement, infrastructure-gap reduction, budget efficiency and sustainability.

11. Example Scenario

Village Budget: ■50 lakh
The system identifies poor drainage in Zone A and poor road connectivity in Zone B.

A user can create:

Plan A: Road ■20 lakh + Drainage ■15 lakh + Water infrastructure ■15 lakh
Plan B: Major road upgrade ■35 lakh + Drainage ■15 lakh
Plan C: AI-optimized mixed intervention

The platform compares outcomes rather than simply declaring one plan universally correct. A plan might improve accessibility significantly but benefit fewer people, while another might reduce infrastructure inequality more effectively.

12. Recommended Technology Architecture

Layer	Recommended Direction
Frontend	React or Next.js
UI/UX	Interactive planning dashboard and scenario controls
2D Mapping	Leaflet, MapLibre or equivalent
3D Visualization	Three.js or equivalent web-based 3D technology
Backend	Node.js API/application layer
Spatial Processing	Python-based geospatial processing where useful
Database	PostgreSQL + PostGIS
Algorithms	C++ or Python depending on integration requirements
Computer Vision	Optional pretrained/fine-tuned segmentation or feature extraction pipeline
Deployment	Cloud/web deployment suitable for live demonstration

13. Suggested Team Roles

Team Capability	Primary Responsibility
System architecture + backend	Overall architecture, APIs, database and integration
C++ + basic DSA	Graph algorithms, accessibility analysis and optimization
Node.js + Python + CV + basic ML	Geospatial analytics, optional CV/ML pipeline and AI modules
3D / Gazebo / prediction systems	3D digital twin, simulation logic and prediction-oriented modules
UI/UX + frontend/vibe coding	Product design, frontend and interactive planning experience
Presentation/content/communication	Research, documentation, PPT, demo narrative, video and final presentation

14. Ruthless MVP Scope

The biggest danger is trying to build a complete national rural-planning platform. Do not do that.

The MVP should demonstrate one village or representative sample dataset and execute one complete end-to-end workflow:

- Load one village dataset
- Display roads and buildings on an interactive map
- Generate a basic 3D view
- Calculate 2–3 infrastructure/accessibility indicators
- Allow the user to add at least two intervention types
- Apply a fixed budget constraint
- Recalculate impact after changes

• Compare at least two scenarios

• Provide one optimization or AI-assisted recommendation

Everything else should be treated as an enhancement, not a dependency.

15. Potential Innovation / USP

The strongest differentiator is the combination of **digital twin + participatory planning + simulation + optimization + gamified scenario design**.

Existing geospatial platforms can visualize data. GramVerse's proposed contribution is to make planning interactive: users can ask *'What happens if we spend the budget this way instead?'* and receive a measurable, visual answer.

16. Major Risks and How to Handle Them

Data access: Do not depend on obtaining a specific restricted dataset. Build a flexible import pipeline and use representative public/sample geospatial data for the prototype.

Overbuilding 3D: A beautiful 3D village without planning intelligence is only a visualization demo. Prioritize the decision/simulation engine.

Fake AI: Avoid adding a chatbot merely to claim AI. Use measurable analytics, CV or optimization where justified.

Too many features: Complete one planning loop end-to-end before adding advanced modules.

Unclear scoring: Every score must be explainable: show the indicators and weights that contributed to the result.

17. Final Pitch

> **GramVerse AI transforms static drone-survey and GIS information into an interactive 3D village digital twin where citizens and planners can experiment with development decisions before they are implemented in the real world. Users work under realistic budget and infrastructure constraints, simulate interventions such as roads and drainage, measure accessibility and community impact, compare multiple development scenarios and receive data-driven recommendations for more effective resource allocation.**

Why This Can Work in SIH

• Direct alignment with the stated requirements of SIH1704

• Strong visual demonstration potential

• Uses the team's combined strengths instead of forcing everyone into unrelated modules

• Has genuine technical depth through GIS, graph algorithms, optimization and optional computer vision

• Can be demonstrated with a realistic MVP without needing a massive production-scale dataset

• Creates a memorable narrative: 'Test development decisions digitally before building them physically.'

Recommended Next Step

Before coding, the team should create a precise MVP specification: exact user roles, one representative village dataset, the first three metrics, two intervention types, one optimization objective, the screen flow, API architecture and a task breakdown. That is the point where we can determine whether GramVerse should be your final SIH selection or whether another statement from the two PDFs has a better execution-to-impact ratio.